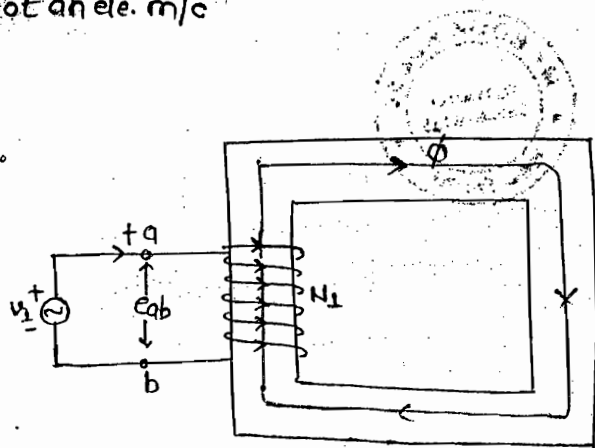


DATE 03/11/19

Transformers

Sagar Sen
887143336

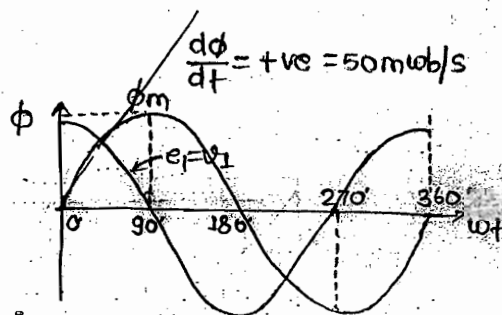
* TF is not an ele. m/c



* Core provides — Low reluctance path
— mech. support

$$\phi = \phi_m \sin \omega t$$

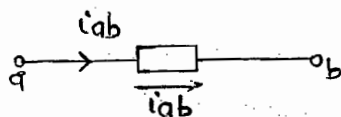
(time expⁿ)



$$\text{Flux linkage } (l) = NI$$

* Whenever there is a flux linkage there is a flux.

V_{ab} — vol. of a wrt b



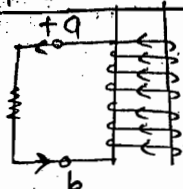
$$e_{ab} = \pm \frac{dl}{dt}$$

* Lenz law → According to this law the dirⁿ of induced emf is such that it is allowed to cause a current (by short circuiting the coil). then the current so produced would have an effect that opposes the cause.

$$\text{Thus } e = \pm \frac{dl}{dt}$$

Where the sign depends on lenz law & which terminal is taken as +ve.

Testing of Faraday's law →

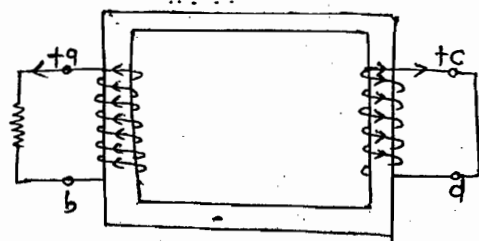


$$e_{ab} = + \frac{d\lambda_1}{dt}$$

$$= + N_1 \frac{d\phi}{dt}$$

$$= 100 \times (50 \times 10^{-3}) \text{ volts}$$

$$= 5 \text{ volts}$$

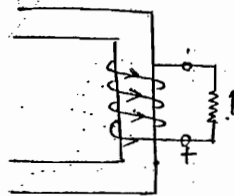
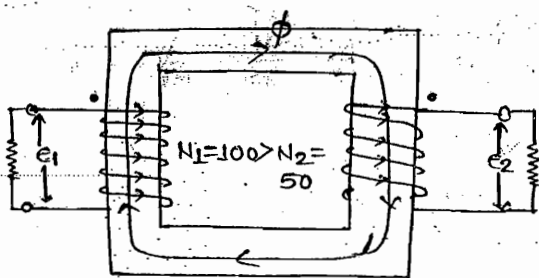


$$e_{cd} = \pm \frac{d\lambda_2}{dt}$$

$$= + \frac{d\lambda_2}{dt}$$

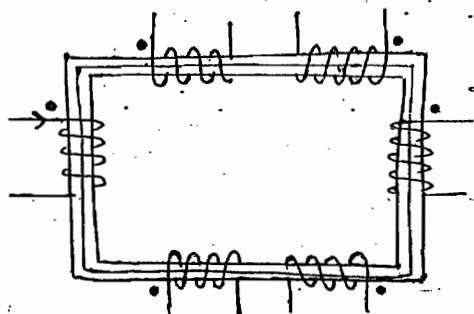
$$= + 50 \times (50 \times 10^{-3})$$

$$= + 2.5 \text{ volts}$$



Dot convention → If the currents enter (or) leave through the dots simultaneously then the fluxes are additive.

* Only the 1st dot is assigned. The remaining dots follow automatically depending upon the sense of winding.



* As applied to Xmer therefore if the currents enter through the dot in the 1st wdg, then it should leave through the dot from 2nd wdg in order to satisfy the lenz law.

In other words the dots have the same instantaneous polarity.

$$\phi = \phi_m \sin \omega t = \phi_m \cos(\omega t + 90^\circ)$$

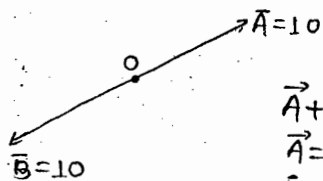
$$e_1 = N_1 \frac{d\phi}{dt}$$

$$= N_1 \frac{d}{dt} (\phi_m \sin \omega t)$$

$$= N_1 \phi_m \omega \cos \omega t$$

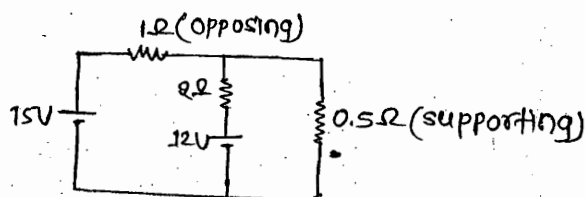
$$\Downarrow$$

$$\sin(\omega t + 90^\circ)$$

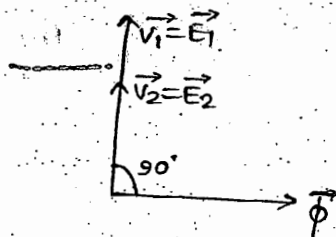


$$\vec{A} + \vec{B} = 0$$

$$\vec{A} = -\vec{B}$$



* Phasor diagram → This diagram must come with circuit diagram.



$$R_{mSE_1} = \frac{N_1 \phi_m \omega}{\sqrt{2}} = \frac{N_1 \phi_m (2\pi f)}{\sqrt{2}}$$

$$E_1 = \sqrt{2} \pi f \phi_m N_1$$

$$e_2 = N_2 \frac{d\phi}{dt}$$

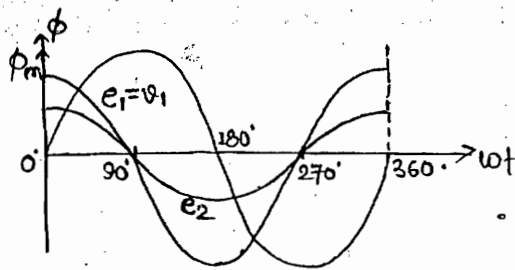
$$= N_2 \frac{d(\phi_m \sin \omega t)}{dt}$$

$$= N_2 \phi_m \omega \cos \omega t$$

$$= N_2 \phi_m \omega \sin(\omega t + 90^\circ)$$

$$E_2 = \frac{N_2 \phi_m \omega}{\sqrt{2}} = \frac{N_2 \phi_m (2\pi f)}{\sqrt{2}}$$

$$E_2 = \sqrt{2} \pi f \phi_m N_2$$



* Ideal transformer $\rightarrow$ (No losses, $\mu = \infty$)

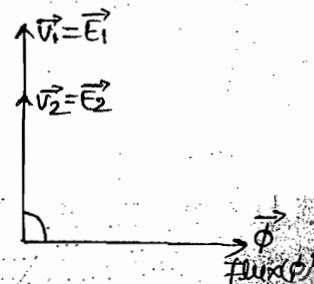
* (flux will establish without any exciting current)

$$\phi = \frac{\text{MMF}}{\text{Reluctance}}$$

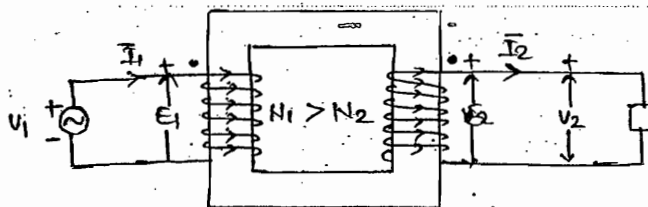
$$\phi = \frac{NI_0}{\frac{l}{\mu_r \mu_0}}$$

$$\phi \times \frac{l}{\mu A} = NI_0$$

$$I_0 = \phi \times \frac{l}{\mu A} \times \frac{1}{N} \quad * (\mu = \infty, I_0 = 0)$$



Phasor dia. of ideal Xmer on NL



MMF Balance on load of ideal transformer

$$N_1 \vec{I}_1 - N_2 \vec{I}_2 = 0$$

$$N_1 \vec{I}_1 = N_2 \vec{I}_2$$

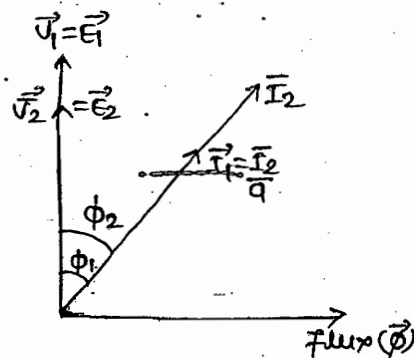
$$\vec{I}_1 = \frac{N_2}{N_1} \vec{I}_2$$

$$\boxed{I_1 = \frac{I_2}{a}}$$

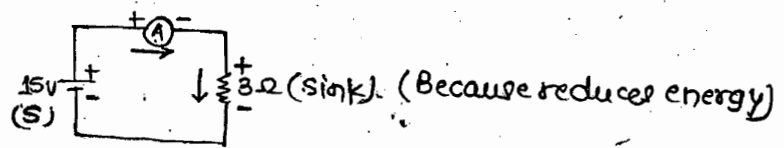
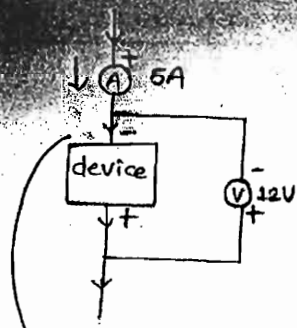
$$\boxed{I_1 = I_2}$$

$$\frac{V_1}{V_2} = \frac{E_1}{E_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1} = a \rightarrow \text{Turn Ratio}$$

I_2' = 2° current referred to 1°



Phasor dia. of an ideal TF on lagging PF load.



Current enter at lower potential & exit from higher potential. Then the device is Source.

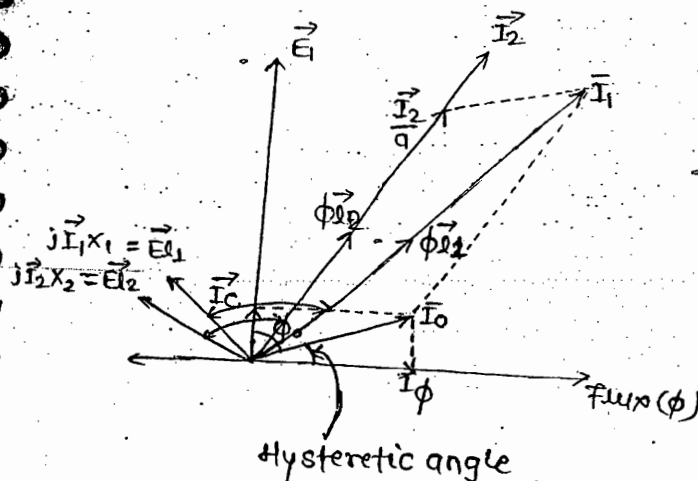
$$\frac{V_1}{V_2} = \frac{E_1}{E_2} = \frac{N_1}{N_2} = a = \frac{I_2}{I_1} = \frac{I_2^*}{I_1^*}$$

$$V_1 I_1^* = V_2 I_2^*$$

$$\& E_1 I_1^* = E_2 I_2^*$$

angle

Hysteretic current → It is the angle b/n sinusoidal flux & sinusoidal current



I_c = core loss component of exciting current (I_0)

I_ϕ = Magnetising current of exciting current

$$I_c = I_0 \cos \phi_0$$

$$I_\phi = I_0 \sin \phi_0$$

$$(\phi_0 = 80^\circ)$$

E_{l1} = 1° leakage flux voltage

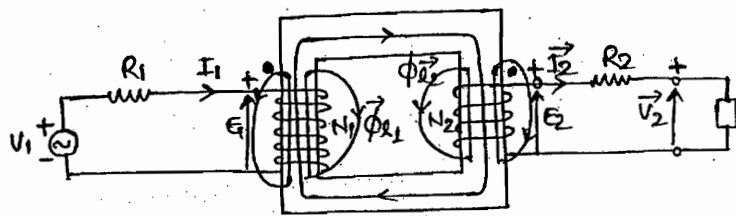
E_{l2} = 2° leakage flux voltage

$$X_2 = \frac{E_{l2}}{I_2}, \quad X_1 = \frac{E_{l1}}{I_1}$$

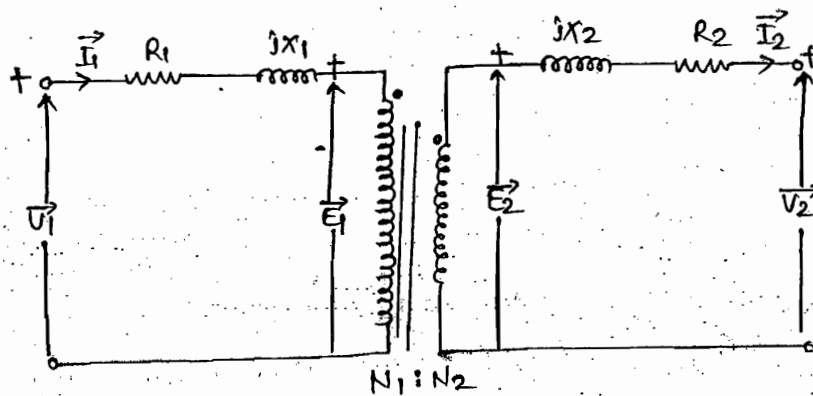
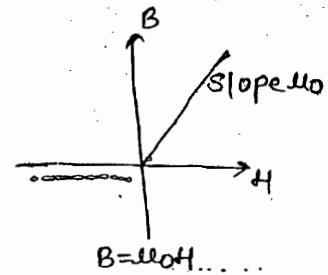
MMF balance of practical TF →

$$N_1 \vec{I}_1 = N_2 \vec{I}_2 = N_1 \vec{I}_0$$

$$\vec{I}_1 = \frac{\vec{I}_2}{a} + \vec{I}_0$$



$\phi_l = \text{leakage flux.}$

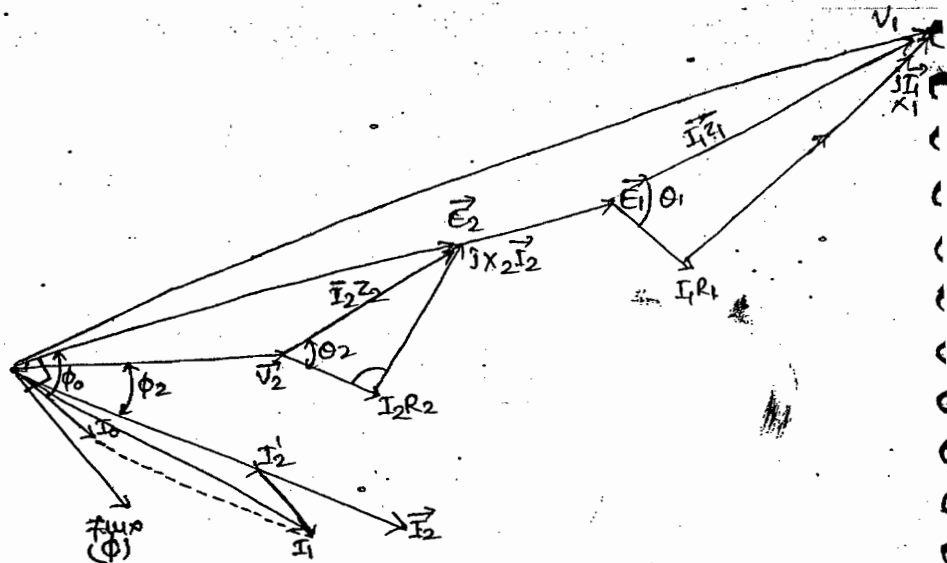


Representation of a practical 2wdg TF

$$\vec{E}_2 = \vec{V}_2 + \vec{I}_2 R_2 + j \vec{I}_2 X_2 \quad \vec{I}_1 = \frac{\vec{I}_2}{a} + \vec{I}_0 \quad \vec{V}_1 = \vec{E}_1 + \vec{I}_1 R_1 + j \vec{I}_1 X_1$$

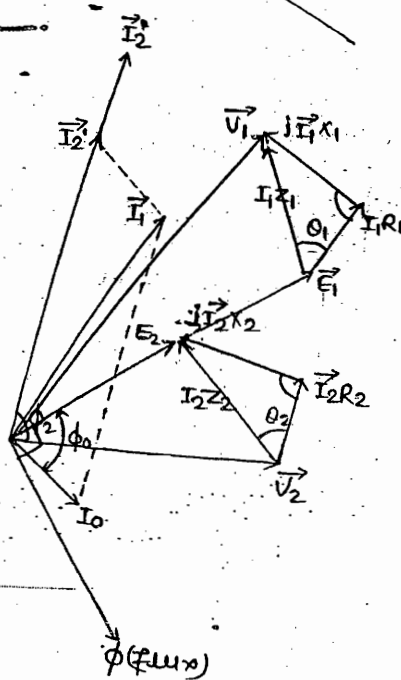
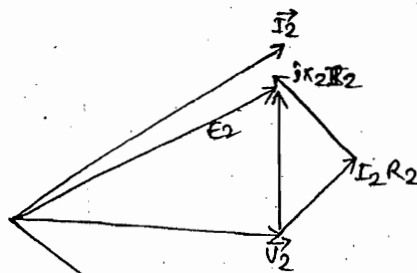
$$\vec{E}_1 = a \vec{E}_2 \quad = \vec{I}_2 + \vec{I}_0$$

Complete phasor diagram $\rightarrow N_1 > N_2$, lagging PF operation.



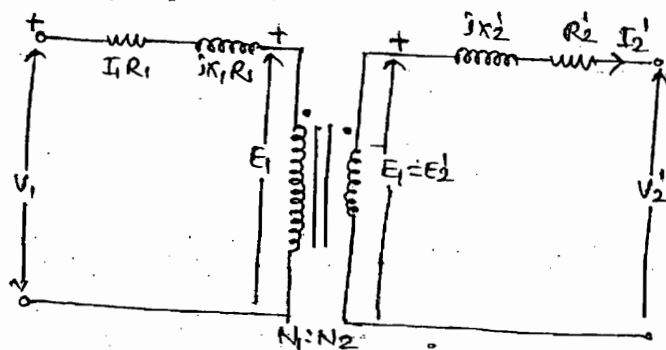
Tight coupling by iron core.

Phasor diagram of operation →



Equivalent circuit → Representation of any device with the help of passive & active elements devices that can be used to analysis & predict the performance of device is called its equivalent circuit.

DATE-04/11/14 (Referred to I_1')



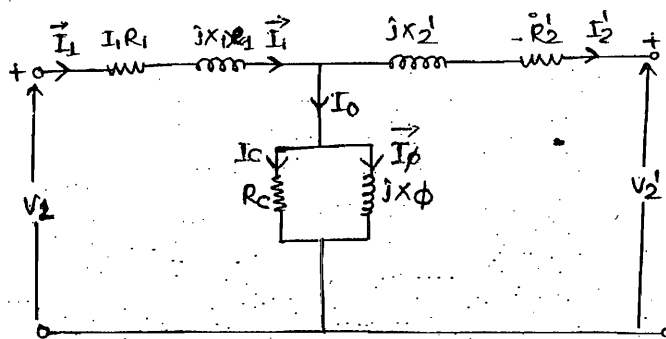
$$\vec{E}_2 = \vec{V}_2 + \vec{I}_2 R_2 + jX_2 \vec{I}_2$$

multiplying by $a = \frac{N_1}{N_2}$

$$a\vec{E}_2 = a\vec{V}_2 + a\vec{I}_2 R_2 + j a\vec{I}_2 X_2$$

$$\vec{E}_2' = \vec{V}_2' + \left(\frac{I_2}{a}\right) \times a^2 R_2 + j \left(\frac{I_2}{a}\right) a^2 X_2$$

$$\vec{E}_1 = \vec{E}_2' = \vec{V}_2' + \vec{I}_2' R_2' + j \vec{I}_2' X_2'$$



R_c = Core loss eq. resistance.

Fig → Exact equivalent ckt
"T" Equivalent

Que → The parameters of a eq. ckt of a 1φ is 150kVA, 2400V/240V Xmer are

$$R_1 = 0.2\Omega, X_1 = 0.45\Omega, R_2 = 2m\Omega$$

$$X_2 = 4.5m\Omega, R_c = 10k\Omega, X_\phi = 1.55k\Omega$$

Using this ckt referred to 1°, determine the 1° i/p vol., i/p current & i/p PF of the Xmer operating at rated load with 0.8 lagging PF.

Soln → FL must be delivered (Rated kVA) at the cost of rated voltage & rated current both.

Rated voltage - Rated Flux

$$1000W, 200V$$

$$180V$$

$$\text{then } 1000 \times \left(\frac{180}{200}\right)^2 = 810W$$

taking V_2 as ref.

$$V_2 = 240\angle 0^\circ ; I_2 = \frac{150 \times 10^3}{240} \angle -\cos^{-1}(0.8)$$

$$= 625 \angle -36.87^\circ A$$

$$V_2' = aV_2 = 10 \times 240 \angle 0^\circ$$

$$a = \frac{2400}{240} = 10$$

$$V_2' = 2400 \angle 0^\circ$$

$$I_2' = \frac{I_2}{a} = \frac{625 \angle -36.86^\circ}{10}$$

$$= 62.5 \angle -36.86^\circ \text{ A}$$

$$Z_2' = a^2 Z_2$$

$$= (10)^2 [(2 + j4.5) \times 10^{-3}]$$

$$= (0.2 + j0.45) \Omega$$

$$E_1 = E_2' = V_2' + Z_2' I_2'$$

$$E_1 = 2400 \angle 0^\circ + 62.5 \angle -36.86^\circ \times (0.2 + j0.45)$$

$$E_1 = 2426.92 \angle 0.35^\circ$$

$$I_\phi = \frac{E_1}{R_C} = \frac{2426.92 \angle 0.35^\circ}{10 \text{ k}} = 0.2427 \angle 0.35^\circ \text{ A}$$

$$I_\phi = \frac{E_1}{jX_\phi} = \frac{2426.92 \angle 0.35^\circ}{j1.55 \times 10^3} \text{ A} = 1.56 \angle -89.64^\circ \text{ A}$$

$$I_0 = -I_C + I_\phi$$

$$= 0.2427 \angle 0.35^\circ + 1.56 \angle -89.64^\circ$$

$$I_0 = 1.5845 \angle -80.84^\circ \text{ A}$$

$$\text{No load PF angle} \rightarrow 80.84 + 0.35 = 81.19^\circ$$

$$\frac{I_\phi}{I_0} = 0.98 \angle -8.8^\circ$$

$$I_1 = I_2' + I_0$$

$$= 62.5 \angle -36.86^\circ + 1.5845 \angle -80.84^\circ$$

$$I_1 = 63.65 \angle -37.86^\circ \text{ A}$$

$$V_1 = E_1 + I_1 Z_1$$

$$= 2426.92 \angle 0.35^\circ + 63.65 \angle -37.86^\circ \times (0.2 + j0.45)$$

$$V_1 = 2454.68 \angle 0.691^\circ$$

$$\text{I/p PF} = \text{Angle of } V_1 - \text{angle of } I_1$$

$$= 0.691 - (-37.86^\circ)$$

$$= 38.55^\circ$$

$$\text{PF} = \cos(38.55) = 0.7821 \text{ lagging}$$

$$\text{PF} = 0.7821 (\text{lag})$$

* FIRST approximate equivalent circuit →

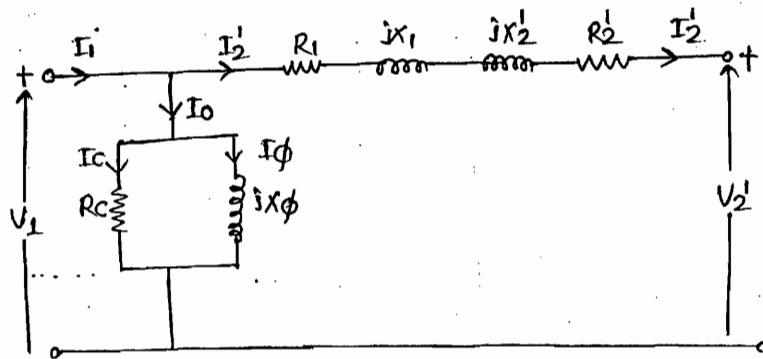
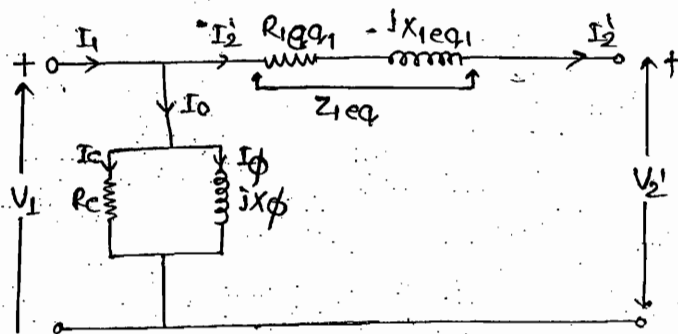


Fig- Cantilever eq:



For η calcⁿ it is best suitable

$$V_1 = 2400 \angle 0^\circ + 62.5 \angle -36.87^\circ \times (0.4 + 0.9j)$$

$$= V_2' + I_2' (Z_1 + Z_2')$$

$$V_1 = 2453.66 \angle 0.69^\circ \text{ V}$$

$$\boxed{V_1 = 2453.66 \angle 0.69^\circ \text{ V}}$$

* 2nd & Final approx eq: ckt →

